

From Represented Information to Realizable Neural Growth

Observation Limits, Singular Positive Splits, and Finite-Sample Tests

Wenyu Huang
University of California San Diego

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Abstract

Can information about a learning unit guide its structural adaptation? We distinguish information retained by a representation, information observed by a controller, and improvement accessible to a specified neural operation. An explicit full-support four-input tanh classifier is a population local minimum while retaining no task information. Two tasks have identical activation-label-residual laws but require different split directions: no passive controller restricted to that law can select an immediate split with positive expected loss reduction on both tasks, even with unlimited samples. Input-residual context permits strict descent, with matching sample order in a two-task identification problem. A separate noisy-parity construction has positive task information and zero splitting matrix. At the same two-child width and positive normalized output weights, fixed-proportion centered splits increase log loss at order t^4 , whereas a child proportion of order t^{d-4} yields a strictly decreasing path at order t^{2d-4} . This is a constrained nonuniform-limit calculation; curvature splitting, parity blindness, and boundary-weight escape have prior precedents. Actual finite-sample tanh training tests the observation bridge against random growth, continued training, fixed-width models, and an explicitly privileged oracle. Geometry helps at a designed initial state under short training budgets; that benefit erodes with longer training and altered initialization. Information-score experiments and holdout gates expose additional estimation and optimization failures. The results establish specific limits and a restricted trainable bridge, not a universal information-driven growth algorithm.

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1 The scientific question after Phase I

The motivating question is whether a learning system can use what its units represent or process to decide autonomously how its structure should change. Activation entropy and neuron splitting are candidate ingredients, not scientific commitments. A useful answer must identify both the information available to the controller and the operations its learning system can perform.

The preserved Phase I manuscript, *The Price of Irreversible Representation Growth: Sharp Four-State Obstructions under Logarithmic Loss*, studies a different object: finite encoders under hard alphabet budgets. It proves a sharp obstruction to simultaneously competitive nested encoders. Its four-state construction and reassignment calculation do not identify states with neurons or bound neural optimization cost. We use its warning about restricted operations as a diagnostic, and move to explicit neural parameters. The new theory does not depend on the unreviewed information- bottleneck onset draft preserved with that project.

The central advance is a pair of inspectable separations and their learning tests. First, a summary can be perfectly estimated yet omit the pairing between residuals and inputs that selects a useful operation. Second, even with that pairing known, structural accessibility depends on the way output weights and incoming perturbations approach a growth event. These claims are more specific than saying that entropy is insufficient. They give an exact observation model, strict population descent alternatives, and failure tests for an actual parameterized learner.

We intentionally separate three questions. Does a representation contain information predictive of the target? Can a controller distinguish which structural operation helps from its observations? Does the allowed operation, followed by the implemented optimizer and data budget, realize the benefit? The experiments do not promote a positive answer to one question into a positive answer to the others.

1.1 What was explored and why this direction was selected

Independent investigations began with the broad original question. They considered finite-state reassignment, information-bottleneck birth geometry, conditional-information operation selection, neural splitting, and local observational limitations. Extending the Phase I source-mass rate remained mathematically interesting but would retain the same rare-state population abstraction. An exact conditional-information rule for Bayes-refitted cells would have a direct decision-tree interpretation and insufficient neural content. Re-deriving negative-curvature growth would duplicate established methods. The chosen route instead exposes a missing premise in transferring information criteria to neural operations, and supplies actual tests of the most direct existing geometric remedy.

The observation example would be undermined by a controller identifying both tasks using only the stipulated identical observation laws. The high-order result would be undermined by a negative fixed-proportion small-radius path under its stated restrictions or a failure of the joint-limit coefficient. The empirical hypothesis that task-information scores select useful neural operations is undermined when equal-size, equal-training comparisons favor random or fixed-width alternatives. Those counterexamples and negative comparisons were actively retained.

2 Starting-point audit and evidence types

All supplied Phase I sources, results, and the authoritative PDF were preserved before modifications in a 63-file ZIP snapshot with per-file SHA-256 hashes. The original project has no supplied Git revision; the hash manifest is the starting identifier. Original-code evaluations at $L = 16, 64, 256, 2048$ match eight stored quantities at all 60 saved digits. An independent

implementation enumerates function tables and computes risk from joint probabilities rather than using the original partition recursion and entropy-difference routine. It recovers 15 partitions and 18 compatible chains; nine compared quantities agree to relative tolerance 10^{-58} . The deterministic total-risk prices are respectively 2.191327, 4.641185, 9.499428, and 27.125385. The audited core proofs revealed no substantive mathematical error. This is an audit of the claims used here, not a new formal certification of every Phase I result.

There are three distinct kinds of Phase II evidence. Theorems below are population statements with full proofs. Exhaustive population calculations check finite neural operations and asymptotic coefficients. Finite-sample experiments fit neural parameters or estimate controller statistics from sampled labels; the evaluator may integrate against the known synthetic law, but those population probabilities are withheld from implemented controllers. Using such an evaluator does not convert finite-sample fitting into oracle learning. Likewise the high-order construction is an oracle existence path, not a trained adaptive mechanism.

3 Nearest results and the bounded novelty claim

Splitting steepest descent already derives a residual-weighted neural splitting matrix and negative-eigenvalue binary growth [7, Sections 2.1–2.3, Theorems 2.2–2.4]. Signed splitting optimizes the analogous second-order form with signed normalized coefficients [9, Theorems 2.1–2.3]; Firefly studies broader function-space growth neighborhoods and optimized candidate additions [8, Sections 2.1–2.3]. The controller used in our curvature experiment is an explicit implementation of the established splitting principle. It is not a new entropy-based algorithm.

The closest boundary result is Ding, Li, and Sun’s stationary-plateau analysis [3]. Their Theorem 3 and its proof in Section 6.3.1 cover locally effective neurons with zero inner Hessian under square loss, using a dormant child’s small outgoing coefficient and nearby input weight. Thus neither higher-order effectiveness nor escape through zero output weight is first established here. Our additional calculation enforces strictly positive normalized weights at every nonzero radius, two children, an exactly conserved weighted incoming mean, and logistic loss; it identifies the explicit parity-dependent weight and risk orders, against an interior-proportion obstruction. We found no equivalent statement with this combination in the inspected sources, but do not claim an exhaustive historical search.

The need to specify local observations and feedback has long been emphasized in local-learning theory [1, Sections 6.4 and 7.1]. Information-theoretic neuron learning also has concrete precedents: infomorphic networks use richer presynaptic and context signals [4]. Our activation-summary restriction excludes that richer input access. It is not an impossibility of local synaptic learning or a claim of the first information-theoretic neural implementation.

Parity orthogonality and gradient blindness are established themes [5, Section 2]. More recent work proves hardness for fixed parities in specific ReLU/perturbed-gradient settings [6], while learnability can change under specially structured distributions [2]. Our fixed-dimension Taylor calculation does not claim new general parity hardness. The moment identities, testing inequalities, and log-loss identities in the proofs are standard tools; the contribution is their exact constrained neural witnesses and the accompanying evidence. A source-by-source record, versions, inspected locations, and unresolved novelty risks are supplied in `research/prior_art.md`.

4 What an activation-only controller cannot observe

We now study actual neural parameters. This model has no discrete representation-cardinality constraint. Its purpose is to separate an information summary from the information needed to realize a structural operation. The observation restriction below is explicit: a physical unit may

also have access to presynaptic inputs, in which case the negative result does not apply.

Let X be uniform on $\{e_1, -e_1, e_2, -e_2\} \subset \mathbb{R}^2$. For an unknown task $s \in \{-1, +1\}$ and $0 < \eta < 1/2$, set

$$q_s(\pm e_1) = \mathbb{P}_s(Y = 1 \mid X = \pm e_1) = \frac{1}{2} + s\eta, \quad q_s(\pm e_2) = \frac{1}{2} - s\eta.$$

Every source-label pair has positive probability. Write $\phi(t) = \tanh t$ and use binary log loss $\ell(f, y) = \log(1 + e^f) - yf$. A one-hidden-unit network has logit $f(x) = c + a\phi(b + w^\top x)$. Fix $b = b_0 > 0$, $a = 1$, $w = 0$, and $c = -\phi(b_0)$, so that $f_0 = 0$ and the predicted probability is $1/2$. At this point the activation is $A = \phi(b_0)$ and the output residual is $R = 1/2 - Y$. All logs below are natural.

Proposition 4.1 (A parameter local minimum with unrepresented task information). *If*

$$0 < \eta < \frac{\phi'(b_0)^2}{4|\phi''(b_0)|}, \quad (1)$$

the stated one-unit parameter vector is a local minimum of population log loss on either task. It lies on a manifold of equivalent constant predictors. Nevertheless

$$I_s(X; Y) = \log 2 - h(\frac{1}{2} + \eta) > 0, \quad I_s(A; Y) = 0.$$

The entire laws of (A, Y, R) are identical for the two tasks.

Proof. Introduce the smooth invertible coordinate $r = c + a\phi(b)$, retaining a, b, w . Then $f(x) = r + a\{\phi(b + w^\top x) - \phi(b)\}$. For every a, b near $1, b_0$, the risk as a function of (r, w) is stationary at $(0, 0)$: its r derivative is $\mathbb{E}_s(1/2 - Y) = 0$, and its w derivative is $a\phi'(b)\mathbb{E}_s[(1/2 - Y)X] = 0$ by sign symmetry of each axis. At this point its Hessian in the (r, w) coordinates is block diagonal, with rr entry $1/4$ and ww block

$$\frac{a^2\phi'(b)^2}{8}I_2 + a\phi''(b) \begin{pmatrix} -s\eta/2 & 0 \\ 0 & s\eta/2 \end{pmatrix}. \quad (2)$$

Indeed $\ell_f = 1/2 - Y$, $\ell_{ff} = 1/4$ at zero, $\mathbb{E}XX^\top = I_2/2$, and the mixed rw entries vanish because $\mathbb{E}X = 0$. Under (1) this Hessian is positive definite at $(a, b) = (1, b_0)$. By continuity there is a compact neighborhood of $(1, b_0, 0, 0)$ and $m > 0$ on which the (r, w) Hessian is bounded below by mI_3 . Taylor's integral formula along the line from $(0, 0)$ to (r, w) , with a, b fixed, therefore gives

$$L_s(r, w; a, b) - \log 2 \geq \frac{m}{2}(r^2 + \|w\|^2)$$

throughout a smaller neighborhood. At $r = w = 0$ the risk remains $\log 2$ for all nearby a, b . This proves a local minimum in all original parameters, allowing its neutral directions.

The marginal law of Y is Bernoulli($1/2$) and every input has conditional entropy $h(1/2 + \eta)$; the information formula follows. Since A is constant, and R is a function of Y , the final assertions follow as well. $\square$

Replace the hidden unit by two units of output weight $1/2$, biases b_0 , and incoming weights $+\epsilon v, -\epsilon v$, retaining the intercept $-\phi(b_0)$, where $\|v\| = 1$. The resulting logit is

$$f_{\epsilon, v}(x) = \frac{\phi(b_0 + \epsilon v^\top x) + \phi(b_0 - \epsilon v^\top x)}{2} - \phi(b_0). \quad (3)$$

At $\epsilon = 0$ this is a function-preserving duplication; at positive ϵ it is the specified structural perturbation. This familiar symmetric split is the operation studied in splitting steepest descent; the score derived below is not claimed as new.

For $t \geq 0$ put

$$d(t) = \phi(b_0) - \frac{\phi(b_0 + t) + \phi(b_0 - t)}{2} = \frac{z(1 - z^2) \tanh^2 t}{1 - z^2 \tanh^2 t}, \quad z = \tanh b_0. \quad (4)$$

Thus $d(0) = 0$ and $0 < d(t) < z$ for $t > 0$.

Theorem 4.2 (Exact observation-channel obstruction and attainable descent). *Let $\Delta_s(\epsilon, v) = L_s(f_{\epsilon, v}) - \log 2$, and put $d_j = d(|\epsilon v_j|)$. Then*

$$\Delta_s(\epsilon, v) = \frac{1}{2} \{ \log \cosh(d_1/2) + \log \cosh(d_2/2) \} + \frac{s\eta}{2} (d_1 - d_2). \quad (5)$$

Consequently:

1. Any possibly randomized controller whose entire task-dependent observation consists of the population law of (A, Y, R) , or any number of independent samples from that law, satisfies

$$\max_{s \in \{-1, +1\}} \mathbb{E}_s \Delta_s(\epsilon, v) \geq 0.$$

Here it may select both $\epsilon \geq 0$ and the unit vector v from these observations. Expectations include its observations and randomness; risk is evaluated on a fresh population input. The controller may also decline to split, which gives zero change.

2. A controller with the input-residual second moment $M_s = \mathbb{E}_s[(1/2 - Y)XX^\top]$ can choose $v = e_2$ for $s = +1$ and $v = e_1$ for $s = -1$. For either task its exact gain is

$$G(\epsilon, \eta) = \frac{\eta d(\epsilon)}{2} - \frac{1}{2} \log \cosh\{d(\epsilon)/2\} \geq \frac{\eta d(\epsilon)}{2} - \frac{d(\epsilon)^2}{16} > 0 \quad (6)$$

whenever $0 < d(\epsilon) < 8\eta$.

Proof. The addition formulas for $\tanh(b_0 \pm t)$ give (4); its denominator is positive. At either point on axis j , the split logit is $-d_j$. For any posterior q ,

$$\mathbb{E}[\ell(-d, Y) \mid q] - \log 2 = \log \cosh(d/2) + (q - 1/2)d.$$

Averaging the two axes proves (5).

The two tasks give identical laws of all permitted controller observations. Consequently its joint distribution of (ϵ, v) is the same under both tasks, including when selected from a random sample. Pointwise,

$$\Delta_+(\epsilon, v) + \Delta_-(\epsilon, v) = \log \cosh(d_1/2) + \log \cosh(d_2/2) \geq 0.$$

Taking the common expectation proves the first part. Independent controller randomization cannot change this calculation.

The matrix is $M_s = \text{diag}(-s\eta/2, s\eta/2)$, so it identifies the displayed axis choice. Substituting it in (5) gives the gain. To verify the inequality used in (6), for $u \geq 0$ one has $\tanh u \leq u$, because the derivative of $u - \tanh u$ is $\tanh^2 u \geq 0$ and the value at zero is zero. Integrating gives $\log \cosh u \leq u^2/2$; evenness handles $u < 0$. The strict positivity follows from $d < 8\eta$. $\square$

For orientation, Taylor expansion of (3) yields

$$\Delta_s(\epsilon, v) = \frac{\epsilon^2}{2} v^\top S_s v + O(\epsilon^4), \quad S_s = \phi''(b_0) M_s.$$

This is the established splitting matrix specialized to log loss. Theorem 4.2 adds an exact finite-displacement statement about which observation channels can select its useful direction. It does not require naming M_s an entropy estimator or a unique-information decomposition. In particular, even the scalar $I_s(X; Y)$ is identical across the tasks and does not specify an axis.

Theorem 4.3 (Finite observations: sufficient and necessary sample order). *Suppose the controller receives n independent full pairs (X_i, Y_i) . Define*

$$U_i = (1/2 - Y_i)(X_{i2}^2 - X_{i1}^2), \quad \hat{s} = \text{sign} \left(\sum_{i=1}^n U_i \right),$$

with any fixed rule for a zero sum. Select the axis prescribed in Theorem 4.2 for $\hat{s}$, with fixed $\epsilon > 0$ satisfying $d(\epsilon) < 8\eta$. On either task,

$$\mathbb{P}_s\{\Delta_s = -G(\epsilon, \eta)\} \geq 1 - \exp(-2n\eta^2). \quad (7)$$

Conversely, for any randomized procedure selecting a symmetric split from n full observations, if its probability of strict population descent is at least $1 - \delta$ on each task, where $0 < \delta < 1/4$, then

$$n \geq \frac{\log(1/(4\delta))}{-\log(1 - 4\eta^2)}. \quad (8)$$

For $0 < \eta \leq 1/4$, the right hand side is at least $3 \log(1/(4\delta))/(16\eta^2)$.

Proof. For task s , U_i takes values $\pm 1/2$ and has mean $s\eta$. Hoeffding's inequality for independent variables with range length one therefore gives $\mathbb{P}_s(s \sum_i U_i \leq 0) \leq e^{-2n\eta^2}$. For completeness its exponential bound follows by applying Markov's inequality to $e^{-t \sum_i (sU_i - \eta)}$, using the bounded-variable moment generating function inequality $\mathbb{E}e^{t(V - \mathbb{E}V)} \leq e^{t^2/8}$ for an interval of length one, and minimizing $e^{-tn\eta + nt^2/8}$ at $t = 4\eta$. The moment generating function inequality follows from convexity by bounding e^{tV} by its endpoint chord and maximizing the centered two-point expression (or by twice integrating that expression's second logarithmic derivative, which is a Bernoulli variance at most $1/4$). Correct identification gives exactly the claimed gain.

Any selected split with $\Delta_s < 0$ must, by (5), satisfy $s(d_2 - d_1) > 0$. Therefore a procedure that achieves strict descent with probability at least $1 - \delta$ yields a test for s with each error probability at most δ : report $+1$ when $d_2 > d_1$ and -1 otherwise. Extra independent randomization cannot improve the optimal testing error.

Write P_+, P_- for the two laws of a full pair. Their Hellinger affinity is

$$\rho = \sum_{x,y} \sqrt{P_+(x,y)P_-(x,y)} = \sqrt{1 - 4\eta^2}.$$

The affinity of product laws is ρ^n . For arbitrary finite laws P, Q with affinity a , Cauchy-Schwarz gives

$$\|P - Q\|_{\text{TV}} = \frac{1}{2} \sum_z |\sqrt{P(z)} - \sqrt{Q(z)}|(\sqrt{P(z)} + \sqrt{Q(z)}) \leq \sqrt{1 - a^2}.$$

The sum of type-I and type-II errors of any test is at least $1 - \|P - Q\|_{\text{TV}}$: this follows directly by writing its acceptance probability as a function in $[0, 1]$ and bounding the difference of expectations by total variation. Its equal-prior average error is thus at least

$$\frac{1 - \sqrt{1 - \rho^{2n}}}{2} \geq \frac{\rho^{2n}}{4} = \frac{(1 - 4\eta^2)^n}{4},$$

where the inequality uses $1 - \sqrt{1 - t} = t/(1 + \sqrt{1 - t}) \geq t/2$. Both errors at most δ imply (8). Finally $-\log(1 - u) = \int_0^u (1 - t)^{-1} dt \leq u/(1 - u)$ and $u = 4\eta^2 \leq 1/4$ imply $-\log(1 - 4\eta^2) \leq 16\eta^2/3$. $\square$

The lower bound applies even to a controller given full inputs; the activation-only controller receives no task-identifying information at any sample size. The upper bound uses ordinary independent sampling, not an oracle population covariance. Neither statement predicts the outcome of unrestricted training after the operation. Such training observes input coordinates and can learn distinctions unavailable to the restricted structural controller.

5 Equal width, unequal access: a nonuniform positive split limit

The preceding example concerns missing observations. Even full population information does not make a fixed structural operation effective. We now hold the number of children, positivity and sum of outgoing weights, and weighted incoming mean fixed, but change the relation between the child weight and the size of the incoming perturbation. This is a restricted separation, not a new discovery of parity hardness or boundary-weight escape; see Section 3.

Let $d \geq 6$ be even, X uniform on $\{-1, 1\}^d$, and $\chi(X) = \prod_{j=1}^d X_j$. For $0 < \eta < 1/2$, set

$$\Pr(Y = 1 \mid X = x) = \frac{1}{2} + \eta\chi(x). \quad (9)$$

This law has full support and $I(X; Y) = \log 2 - h(1/2 + \eta) > 0$, independently of d . No limiting rare source mass is used when d is fixed. Fix a smooth activation σ near b with $\sigma''(b) \neq 0$ and $\sigma^{(d)}(b) \neq 0$, with bounded derivatives through order $d + 2$ there. The experiments use $\sigma = \tanh$ and $b = 1/2$. The parent logit is the constant $\sigma(b) - \sigma(b) = 0$. For a unit vector v put $V = v^\top X$, and define a positive binary split

$$f_{t,p}(x) = p\sigma(b + tV) + (1 - p)\sigma\left(b - \frac{p}{1 - p}tV\right) - \sigma(b), \quad 0 < p < 1. \quad (10)$$

There are exactly two hidden units. Their outgoing weights sum to one, their weighted incoming mean is zero, and their outgoing ℓ_1 norm is one. For $p \leq 1/2$, both incoming weights have norm at most t . The intercept stays fixed in this comparison; we do not claim an obstruction after arbitrary intercept refitting or subsequent parameter training.

Write $R(f) = \mathbb{E}[\log(1 + e^{f(X)}) - Yf(X)]$ and

$$K(v) = \frac{\sigma''(b)^2}{32}\mathbb{E}V^4 > 0, \quad B(v) = \sigma^{(d)}(b) \prod_{j=1}^d v_j. \quad (11)$$

A direction with $B(v) > 0$ exists by changing the sign of one nonzero coordinate. Its selection here is an oracle construction.

Theorem 5.1 (A constrained weight–radius separation). *Under the preceding conditions:*

(i) *For every fixed $p \in (0, 1)$ and unit v ,*

$$R(f_{t,p}) - \log 2 = K(v) \left(\frac{p}{1 - p}\right)^2 t^4 + o(t^4) > 0$$

for all sufficiently small positive t . This positivity is uniform when p lies in any compact subinterval of $(0, 1)$ and v ranges over the unit sphere.

(ii) *Choose $B(v) > 0$ and let $p(t) = \kappa t^{d-4}$ for fixed $\kappa > 0$. Then*

$$R(f_{t,p(t)}) - \log 2 = \{K(v)\kappa^2 - \eta B(v)\kappa\}t^{2d-4} + o(t^{2d-4}). \quad (12)$$

In particular, the choice

$$\kappa = \frac{\eta B(v)}{2K(v)}$$

gives strict descent with leading coefficient

$$-\frac{\eta^2 B(v)^2}{4K(v)}.$$

(iii) *At the constant parent, the residual splitting matrix is zero. More generally the derivatives in incoming weights of the label-dependent part of the population loss agree with the independent-label law through every order strictly below d .*

Proof. For every integer $j < d$, $\mathbb{E}[\chi(X)V^j] = 0$. Indeed expand V^j into monomials: at least one coordinate has even exponent (possibly zero), and multiplication by χ makes its expectation zero. At degree d , the only surviving monomials use each coordinate once, so

$$\mathbb{E}[\chi V^d] = d! \prod_i v_i. \quad (13)$$

Odd-degree moments with χ vanish here because d is even and the law is invariant under $X \mapsto -X$. The exact loss identity is

$$R(f) - \log 2 = \mathbb{E} \log \cosh(f/2) - \eta \mathbb{E}[\chi f]. \quad (14)$$

For fixed p , Taylor expansion of (10) cancels the constant and linear terms and gives, uniformly on the finite input set,

$$f_{t,p} = \frac{\sigma''(b)}{2} \frac{p}{1-p} t^2 V^2 + O(t^3).$$

Since $\log \cosh(z/2) = z^2/8 + O(z^4)$, its convexity penalty has leading term $K(v)[p/(1-p)]^2 t^4$. Equations (13) and the Taylor expansion through order d show that $\mathbb{E}[\chi f_{t,p}] = O(t^d)$. Thus the signal is $o(t^4)$. Uniformity on compact p intervals follows from bounded derivatives and $|V| \leq \sqrt{d}$; $\mathbb{E}V^4 \geq (\mathbb{E}V^2)^2 = 1$ ensures a uniform positive leading coefficient.

For the joint limit it is important to retain the factor p in the remainders. Put $q = p/(1-p)$. For $p \leq 1/2$, the two Taylor remainders of order three are bounded by $C[p + (1-p)q^3]t^3 = O(pt^3)$. Consequently, as both $p, t \rightarrow 0$,

$$f_{t,p} = \frac{1}{2} \sigma''(b) p t^2 V^2 \{1 + O(p)\} + O(pt^3),$$

where the braces apply to the displayed leading term, not to a pointwise relative error at $V = 0$. It follows that

$$\mathbb{E} \log \cosh(f_{t,p}/2) = K(v) p^2 t^4 \{1 + O(p) + O(t)\}. \quad (15)$$

Expanding through order $d+1$, all lower-degree terms and the degree- $d+1$ term disappear after multiplying by χ . The remaining Taylor errors are $O(pt^{d+2})$. Hence

$$\begin{aligned} \mathbb{E}[\chi f_{t,p}] &= B(v) t^d \left\{ p + (1-p) \left(\frac{p}{1-p} \right)^d \right\} + O(pt^{d+2}) \\ &= B(v) p t^d \{1 + O(p^{d-1}) + O(t^2)\}, \end{aligned}$$

where the last line uses $B(v) > 0$ fixed. Substitute $p = \kappa t^{d-4}$ into this formula and (15). Both leading terms then have order t^{2d-4} , proving (12). The optimizing coefficient is obtained by completing the square; $p(t) \in (0, 1/2]$ eventually.

Finally, the residual splitting matrix is $-\eta \sigma''(b) \mathbb{E}[\chi X X^\top] = 0$ by the same moment argument. The difference of the parity-task risk and the independent-label risk for any logit f is exactly $-\eta \mathbb{E}[\chi f]$. For a finite sum of smooth neurons with zero incoming weights, any derivative of this difference in incoming coordinates of total order less than d is a linear combination of moments $\mathbb{E}[\chi X^\alpha]$ with $|\alpha| < d$, and vanishes. This proves (iii). It asserts a derivative identity, not an impossibility for arbitrary sample-based algorithms, active probing, or optimization away from the parent. $\square$

The decreasing path approaches an embedding with a zero-output child at $t = 0$. It is not a small perturbation of the equal-weight duplication in the full widened parameter vector: its outgoing proportions differ by order one from $(1/2, 1/2)$. The matched constraints are width, outgoing absolute sum, incoming radius, and weighted incoming mean, not full parameter displacement, initialization, or optimization cost.

The theorem compares positive operations at the same width. It does not say that a vanishing weight is the unique or practical remedy. A fixed nonzero radius, a different activation, a changed intercept, untied retraining, signed coefficients, or a different architecture can alter the conclusion. In particular, the higher-order descent can be extremely small despite a substantial information gap.

Proposition 5.2 (Raw score signal-to-noise cost). *For the same even-parity law, let $H_t(X) = \sigma(b + tV) - \sigma(b) - t\sigma'(b)V$ and estimate $m_t = \mathbb{E}[(Y - 1/2)H_t(X)]$ by an iid sample average. If $B(v) > 0$, then*

$$m_t = \eta B(v)t^d + O(t^{d+2}), \quad \text{Var}((Y - 1/2)H_t) = 2K(v)t^4 + o(t^4).$$

Thus the sample size needed for this estimator's standard error to be at most m_t is asymptotic to $2K(v)[\eta^2 B(v)^2]^{-1}t^{-2d+4}$.

Proof. The mean follows from (13). Also $H_t = \sigma''(b)t^2V^2/2 + O(t^3)$, while $(Y - 1/2)^2 = 1/4$ identically. Its second moment is therefore $\sigma''(b)^2\mathbb{E}V^4t^4/16 + o(t^4) = 2K(v)t^4 + o(t^4)$. Subtracting the squared mean changes only order t^{2d} . For n iid observations the variance of the average is this variance divided by n , which gives the stated standard-error criterion. $\square$

This is an exact asymptotic description of a particular naive score estimator, not an information-theoretic sample lower bound. If the parity feature is known, $\mathbb{E}[(Y - 1/2)\chi(X)] = \eta$ is estimable without this t penalty. Knowing or discovering that feature is a different information resource.

6 Population and finite-observation tests of the singular path

The source law in Section 5 is fixed for each dimension. We enumerate it for $d = 6, 8, 10$, $b = 1/2$, $\eta = 0.15$, and 11 incoming radii from 10^{-4} to 0.3 . One coordinate of $v = (1, \dots, 1)/\sqrt{d}$ is reversed when necessary to make $B(v) > 0$. All 66 operation evaluations are saved at 100-digit precision, and six smallest-radius cases are checked at 170 digits. The independent checker instead enumerates all 2^d inputs, evaluates direct Bernoulli log loss, and obtains tanh derivatives through integer polynomial recurrence. Its 40 cases include a second fixed proportion and coefficients below and above the descent interval. Twelve shared evaluations agree to better than $1.4 \cdot 10^{-60}$ relative error.

For $d = 6, t = 0.1$, balanced splitting changes risk by $+4.2255 \cdot 10^{-6}$ nats; the vanishing proportion $p = 0.00286119$ changes it by $-3.5902 \cdot 10^{-11}$ nats. At the same radius the latter gains for $d = 8, 10$ are approximately $8.47 \cdot 10^{-16}$ and $3.68 \cdot 10^{-20}$ nats. The task-information gap is about 0.04570 nats in every case. Strict descent therefore does not constitute a useful trained classifier. The asymptotic gain is not uniform over radius: for $d = 10, t = 0.3$, the vanishing-proportion formula actually increases risk by $3.46 \cdot 10^{-12}$ nats, a saved finite-radius failure. A coefficient three times the asymptotically minimizing κ reverses the small-radius improvement in the independent check, demonstrating that merely making a child small is insufficient.

We also ran finite-observation tests of Proposition 5.2 for $d = 6$. For each of four radii and five sample sizes, 1,000 independent repetitions estimate the raw score. Sampling uses multinomial counts of the exact iid input-label categories; this is ordinary random sampling represented by sufficient counts, with no importance weighting or population labels in the estimate. A comparator receives the true parity feature and estimates its label correlation on the same samples. That feature is privileged structural information and is explicitly not an equally informed controller.

At $t = 0.1, n = 1,048,576$, the raw score has the wrong or zero sign in 507 of 1,000 repetitions. Its unit-signal-to-noise sample proxy is $1.3844 \cdot 10^{10}$ observations. At $t = 0.4$ the same sample

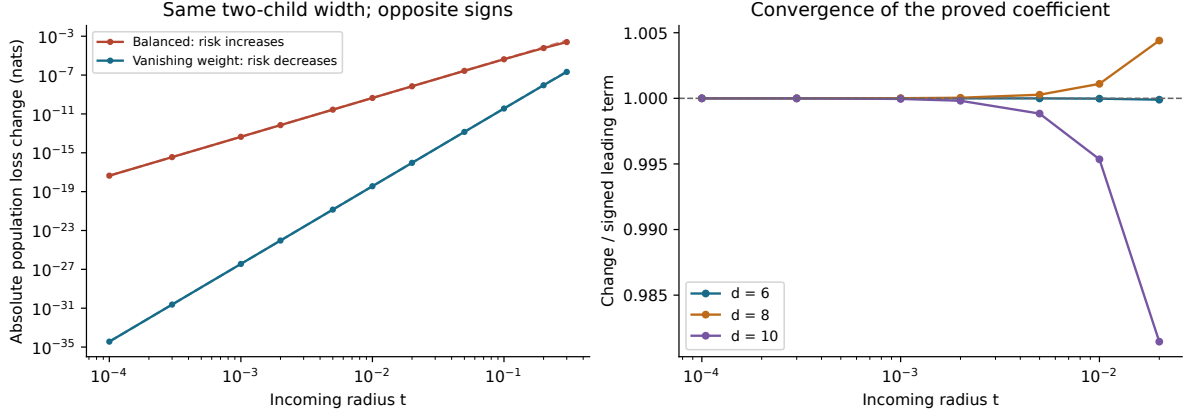


Figure 1: Exact population calculations, not training. Left: absolute loss change for balanced positive splitting (increases) and the vanishing-proportion path (decreases) at $d = 6$. Curves use saved high-precision outputs; dashed curves are proved leading terms. Right: the measured change divided by its signed leading term for $d = 6, 8, 10$. The right panel restricts to smaller radii to show the asymptotic regime. Both operations use two children and outgoing absolute sum one. The diminishing gain is shown explicitly.

size has 18 failures and the proxy is $2.4928 \cdot 10^5$. The supplied parity feature has no observed sign errors across the tested conditions. These measurements show the cost of this raw score, not a minimax lower bound on learning parity, nor a claim that a known feature is free to discover.

7 Finite-sample neural growth: realized effects and failed extensions

The preceding statements concern population risks and prescribed operations. We therefore conducted actual neural training experiments in which the controller observes only sampled labels. These experiments test three different questions: can sampled operational statistics recover the useful split direction; does that initial advantage survive optimization; and can an empirical gate avoid unnecessary growth? A separate exploration tests a conditional-information operation score under ordinary random neural initialization. None of these experiments identifies representation states with neurons.

7.1 Parameterized system and controller information

The trained predictor is

$$p_{\theta}(x) = \text{sigmoid} \left(c + \sum_{j=1}^k a_j \tanh(b_j + w_j^{\top} x) \right).$$

The trainable objects are actual hidden units with input weights w_j , biases b_j , output coefficients a_j , and a common intercept c . A split duplicates unit j , halves its output coefficient, and replaces its two input weights by $w_j + \varepsilon v$ and $w_j - \varepsilon v$. All input weights, hidden biases, output coefficients, and the intercept are subsequently trained without a tying constraint. At $\varepsilon = 0$ this operation preserves the predictor exactly; for positive ε it is a controlled perturbation, followed by a real change in parameter count. Retraining may change predictions at every old input. It need not create a nested discrete partition.

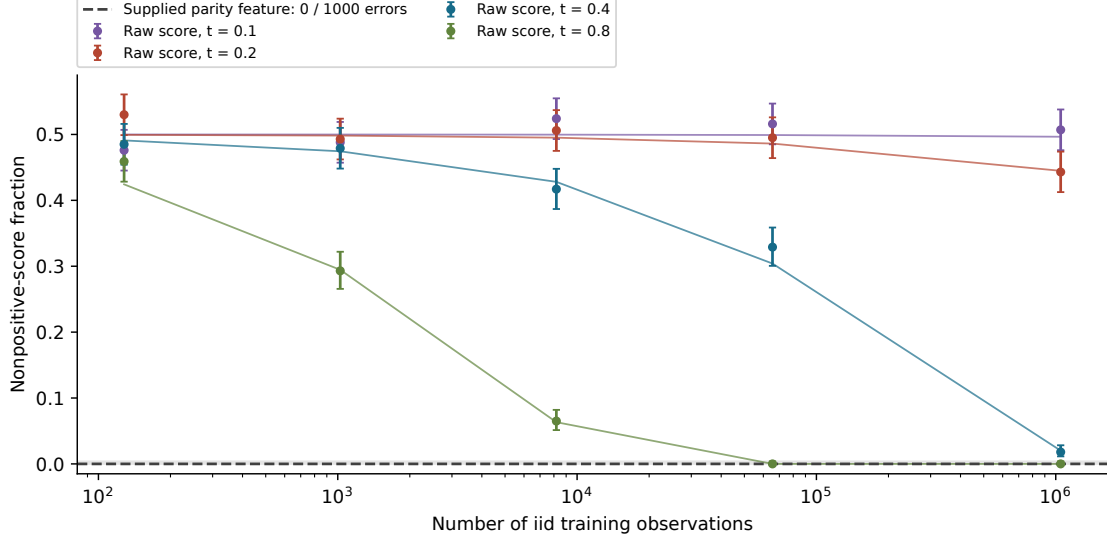


Figure 2: Finite iid score estimation on the six-bit noisy-parity task. Points show the fraction of nonpositive estimated scores across 1,000 repetitions; bars are pointwise 95% Wilson binomial intervals. The population score is positive. Curves are normal-approximation predictions using exact population means and variances, not fitted error rates or guarantees. The dashed reference uses a supplied true-parity feature and has zero observed failures in these runs; its uncertainty remains nonzero. No neural parameters are trained in this figure.

For the theory-linked experiment, the learned direction is the smallest eigenvector of

$$\hat{S}_j = \frac{1}{n} \sum_{i=1}^n [p_\theta(x_i) - y_i] a_j \tanh''(b_j + w_j^\top x_i) x_i x_i^\top.$$

This is an operation-specific curvature statistic, not marginal activation entropy. Its implementation observes input coordinates, residuals, and the current parameters. It receives neither the population conditional label probabilities nor a supplied signal direction. The random-direction control uses a normalized standard Gaussian vector. A separately labelled population curvature oracle computes S_j from the complete known synthetic population; it is a diagnostic and does not enter either learned controller. At the constant start, an activation-only scalar statistic has no directional information; the random control is the deliberately uninformative directional comparison, rather than an invented entropy optimizer.

7.2 The controlled bridge and its validity tests

We use five regimes. The four-state regime is uniform on $\{\pm e_1, \pm e_2\}$ with conditional label probabilities 0.65 on the first axis and 0.35 on the second. The width-one initial parameters are $w = 0$, $b = 0.5$, $a = 1$, and $c = -\tanh(0.5)$, so the initial prediction is $1/2$. The rotated regime is uniform on the sixteen points $\{\pm Qe_j : 1 \leq j \leq 8\}$, where each repetition draws a new orthogonal matrix by QR factorization of a Gaussian matrix. Label probabilities are 0.65 and 0.35 on its first two axes and 0.5 on the remaining axes. Thus the controller must discover the input geometry amid six irrelevant axes. The sample-pretrained regime uses this rotated task but first trains the width-one model for 100 updates from the special initialization. The random-start regime instead initializes all width-one weights randomly and trains for 100 updates. The no-signal regime uses the rotated inputs with every label probability equal to 0.5. These tests distinguish a deliberately designed stationary start from a checkpoint obtained through ordinary sample training.

Each regime uses $n \in \{256, 1024\}$ IID training examples and an independent validation sample of the same size. A sample contains ordinary Bernoulli labels; there is no importance sampling or enumerated-label training. The split uses $\varepsilon = 0.6$. Each branch receives either $T = 30$ or $T = 300$ full-batch Adam updates with learning rate 0.02, standard momentum coefficients 0.9, 0.999, and numerical denominator constant 10^{-8} . Optimizer moments are reset at each branch, including the continuation control. The shared pre-split checkpoint is identical for the spectral, random, and oracle branches. All parameters are untied and trained after the split. The two budgets are separate runs, not selected stopping times. For numerical stability, the code clips predicted probabilities to $[10^{-9}, 1 - 10^{-9}]$ for evaluation; the maximum absolute logit across all saved fitted bridge branches is 6.852, well inside the clipping threshold of 20.723. Thus that clip does not alter their reported population losses.

There are 64 independent repetitions per regime, sample size, and budget, using seeds 27100 through 27163. This gives 1280 completed training configurations. Shared random seeds make within-configuration method comparisons paired; confidence intervals never treat the methods or input points as independent replicates. We report Student- t intervals across the 64 seed-level paired risk differences. They are descriptive intervals for these synthetic regimes, without a multiple-comparison claim. Risk evaluation integrates both the finite input distribution and the conditional label law exactly. This exact evaluation is distinct from the finite-sample learning that produced each network.

The autonomous variant fits both the grown model and a width-one continuation. On the disjoint validation sample it computes their paired per-example log-loss improvements D_i , accepting growth only if $\bar{D} > 1.96 s_D / \sqrt{n}$. Otherwise it keeps the continuation. This fixed normal-approximation rule is a heuristic gate, not a theorem-level finite-sample confidence guarantee. Both spectral and random proposals use the same rule and validation sample. The gate therefore does not silently borrow population information, but it does pay for two fitted branches and additional labelled validation data. Its comparison with an ungated continuation does not by itself isolate the value of the spectral principle.

7.3 Capacity, compute, data, and implementation controls

A width- k model in dimension d has $k(d+2)+1$ trainable parameters. The bridge therefore grows from $d+3$ to $2d+5$ parameters. Spectral and random growth have identical final architecture, shared initialization, training data, update count, and optimizer. Their additional controller costs differ: forming the spectral matrix costs $O(nd^2)$ arithmetic and its eigendecomposition costs $O(d^3)$; random direction generation costs $O(d)$. These overheads are explicitly additional, although small in these low-dimensional experiments.

We also train fixed width-two networks from standard random initialization. One comparison matches the number of optimization updates, including any width-one pretraining. A second matches the approximate number of parameter-example updates: after P pretraining steps and T growth steps, the fixed model receives

$$T + \left\lfloor \frac{P(d+3)}{2d+5} \right\rfloor$$

updates. This is an operation-count proxy, not a claim of identical measured FLOPs, optimizer cost, or wall time. $P = 0$ for the prescribed stationary starts and $P = 100$ for the two pretrained regimes. We do not call equal updates and equal compute the same comparison. All methods use the same training examples; only gated decisions consume the separate validation labels. Raw records retain these resource formulas, controller costs, selected directions, split matrices, immediate and post-training risks, validation events, prediction changes, parameter displacements, and fitted network parameters.

The reported parameter counts describe individual predictors, not total live process memory. A gate retains a continuation and a growth candidate, together with optimizer state during fitting;

Table 1: Actual finite-sample neural training at $n = 1024$. Entries are exact population log loss after fitting; Δ is the paired spectral-minus-random mean with a 95% Student- t interval half-width across 64 independent seeds. T is the post-growth update budget. Fixed width uses the approximate parameter-update budget described in the text. Lower is better.

Regime	T	Spectral	Random	Fixed 2	Continue 1	$\Delta \pm$ half-width
Four states	30	0.65606	0.66809	0.68782	0.69323	-0.012033 ± 0.003104
Four states	300	0.64928	0.64928	0.65256	0.68002	-0.000002 ± 0.000003
Rotated 8D	30	0.69175	0.69458	0.69526	0.69744	-0.002827 ± 0.000588
Rotated 8D	300	0.68955	0.68971	0.69159	0.69488	-0.000162 ± 0.000251
Random start	30	0.69473	0.69450	0.69352	0.69450	$+0.000236 \pm 0.000180$
Random start	300	0.69373	0.69371	0.69124	0.69421	$+0.000019 \pm 0.000609$
No signal	30	0.69866	0.69797	0.69640	0.69770	$+0.000691 \pm 0.000179$
No signal	300	0.70064	0.70068	0.70008	0.69805	-0.000046 ± 0.000064

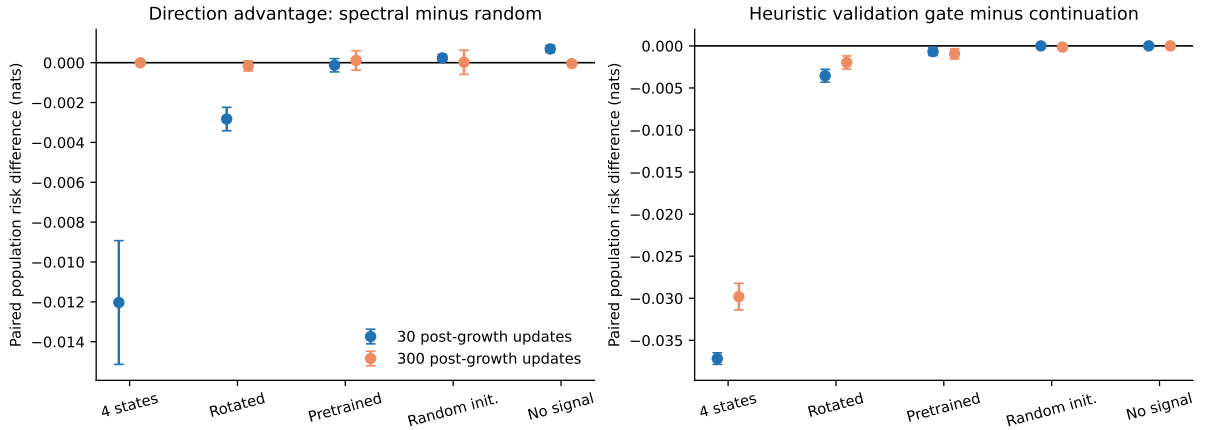


Figure 3: Measured neural training results at $n = 1024$. Left: spectral minus random split risk at identical architecture and training budgets. Right: the heuristic validation-gated spectral model minus width-one continuation; this latter comparison additionally uses validation labels and branch-training compute. Points average 64 paired repetitions; bars are 95% Student- t intervals. Every risk is evaluated exactly on the finite population after fitting from IID labelled samples. Negative values favor the spectral or gated method.

its peak memory and branch-training cost exceed the final predictor size. The implementation additionally fits every comparison and oracle for measurement. Those research costs cannot be credited as free online adaptation. The entire 1280-configuration bridge run took 85.53 seconds on the local CPU environment; this aggregate wall time is not a per-method runtime comparison.

Independent implementation checks compare analytic gradients against central finite differences (maximum absolute error 8.16×10^{-11}), verify exact function preservation at zero split displacement, and compare the actual small-displacement risk change with the derived quadratic term. At $\varepsilon = 10^{-4}$ their ratio is 1.000000199. These checks support the implementation; they do not replace the mathematical proofs.

7.4 What the bridge established, and where it failed

At the prescribed four-state start the learned spectral direction selects the correct axis in all 64 repetitions at both sample sizes. Its immediate population loss reduction is 0.00759820 nats; random directions have negative mean reductions, i.e. they initially increase the loss. At $n = 1024$ and 30 updates, spectral growth has mean risk 0.656061, compared with 0.668094

for random growth. Because those branches share the same capacity, data, training budget, and initial checkpoint, this is evidence for the operational direction choice in the designed regime, subject to its extra matrix-estimation cost. After 300 updates their risks are 0.649278 and 0.649280; the directional advantage has essentially disappeared. Thus the local geometric advantage is not a persistent representational advantage when later optimization can reorganize both models.

On the rotated task, the spectral proposal immediately improves population loss in 44 of 64 repetitions at $n = 256$ and 61 of 64 at $n = 1024$. Its immediate mean gain rises from 0.001245 to 0.001795 nats, compared with 0.001900 for the population curvature oracle. At $n = 1024$ its 30-update advantage over random splitting is 0.002827 nats; it shrinks to 0.000162 after 300 updates. At $n = 256$, longer retraining instead overfits: spectral population risk grows from 0.709556 after 30 updates to 0.722064 after 300, both above the constant-predictor risk $\log 2$. This is not hidden by reporting only ratios or training accuracy.

The sample-pretrained and random-start tests are essential limitations. After ordinary width-one sample training, the selected spectral direction is no longer reliably aligned with the designed signal. In the random-start regime at $n = 1024$, its 30-update risk is slightly higher than random growth (0.694734 versus 0.694498). At 300 updates there is no resolved difference. The experiment therefore does not establish a generally useful growth rule at arbitrary trained checkpoints. Moreover, finite-sample training can move along initially neutral parameter directions and escape the particular population local minimum: width-one continuation is not required to remain constant. The observed continuation improvements do not contradict a local population statement at a specified parameter point.

Every forced spectral split in the no-signal task increases the immediate population loss. At $n = 1024$ the validation gate accepts zero of 64 spectral proposals at both budgets; at $n = 256$ it accepts one of 64 at each budget. On the rotated $n = 256$, 300-update condition it rejects all 64 proposals, preventing selection of the heavily overfit grown networks. Rejection is relative to a fitted continuation, which may itself overfit; the gate does not recover the unknown Bayes predictor or prove universally safe adaptation. These are empirical gate diagnostics, not an automatic conversion of the curvature theorem into a finite-sample acceptance guarantee.

7.5 Independent random-initialization information-score probe

Before choosing the theory-linked experiment, an independent investigation implemented actual width-two to width-three growth on continuous Gaussian inputs. It initially used the conditional label information of the *antisymmetric child-activation contrast* $\tanh(a + \delta u) - \tanh(a - \delta u)$, where $a = b_j + w_j^\top x$ and $u = v^\top x$. Independent adversarial review identified a material limitation: this feature contrast is not the symmetric predictor deformation caused by the prescribed initial split. Its negative result cannot establish failure of an operation-aligned information score. We retain all original runs and add a fresh-seed ablation with the exact initial logit deformation

$$Z_v(x) = a_j \left\{ \frac{\tanh(a + \delta u) + \tanh(a - \delta u)}{2} - \tanh(a) \right\}.$$

The corrected feature definition was fixed before drawing its new seeds.

Each candidate is scored by the plug-in conditional mutual information between its three-bin feature and the binary target, conditional on a three-quantile-bin current prediction. The aligned feature uses three empirical quantile bins. The original antisymmetric feature instead uses the fixed thresholds $-0.08, 0.08$; its marginal entropy is retained as a separate baseline. Thus “entropy” here refers to that specified three-state activation-contrast histogram, not differential entropy. The conditioning variable is a lossy quantization of the current prediction, not the full neural representation. Consequently neither score is automatically the recoverable

predictive information of the full model, and neither provides a Bayes-refitting guarantee for the trained network.

For each run, 512 examples train the network, 512 disjoint labelled examples score candidate operations, and 8192 independent inputs evaluate it using the known teacher label probabilities. This evaluation integrates the conditional Bernoulli label law exactly but only Monte Carlo averages the continuous input distribution; it is not an exact population calculation. Population probabilities are never passed to learned selectors. The deliberately optimistic evaluation oracle chooses the candidate with minimum loss on those same 8192 evaluation inputs, and is labelled as an evaluation oracle rather than an implementable controller or an exact population optimizer.

All three tasks use $X \sim N(0, I_3)$ and $P(Y = 1 | X) = 0.05 + 0.90 \text{sigmoid}(g(X))$. The linear-null task has $g(x) = 2x_1$. The localized task has $g(x) = 1.5x_1 + 4 \tanh(2x_2 + 1) \tanh(2x_3 - 1)$. The teacher task uses four tanh units with fixed weights, biases, and output coefficients specified exactly in the released code. Width-two students start randomly and receive 300 full-batch Adam updates with learning rate 0.025. Each of the two units has four independently drawn Gaussian split directions, normalized to unit length, for eight candidate operations. The perturbation scale is $\delta = 0.25$ and each candidate then receives 150 updates of all parameters. Random choice, the two information scores, and marginal entropy select among the same candidates before retraining. A separately labelled trained-validation control trains all eight candidates and selects by score-set log loss. It has an eightfold candidate-fitting cost; that benefit cannot be credited to a cheap information score. Width-two continuation and fixed width-three controls receive the corresponding prescribed update budgets. Information scoring uses 512 additional labelled score examples; random selection does not use those labels, and the entropy proxy uses only score inputs. Therefore any positive comparison with random selection would also have to account for this extra supervised information. Here no resolved aligned CMI advantage survives even with that access. The fixed-width approximate compute comparator scales the width-two pretraining budget by 2/3, explicitly a leading-width proxy rather than exact parameter/FLOP matching.

The original four-seed pilot used 250 pretraining updates and is preserved; subsequent exploratory and aligned runs use 300. Thus the pilot-to-expanded comparison does not change only the repetition count. The 32-seed exploratory run is also preserved, with seeds 17000, ..., 17031. The aligned ablation uses 32 new seeds 41000, ..., 41031 after the audit-triggered definition change. No fresh evaluation result was used to tune that feature definition, the bin count, the optimizer, or the candidate perturbation. It remains a small synthetic mechanism study, not a broad benchmark or a universal rejection of information-guided growth.

For complete specification, the teacher function is $g(x) = \tanh(x^\top W + \beta^\top)\alpha$ with

$$W = \begin{pmatrix} 1.8 & -1.4 & 0.4 & 1.2 \\ 0.4 & 1.8 & -1.7 & 0.8 \\ -1.5 & 0.3 & 1.3 & -1.2 \end{pmatrix}, \quad \beta = (0.4, -0.5, 0.2, 1)^\top, \quad \alpha = (2, -2, 2, -1.5)^\top.$$

Standard random initialization in both studies uses independent input weights $N(0, 0.8^2)$, hidden biases $N(0, 0.1^2)$, output coefficients $N(0, 0.5^2)$, and intercept zero.

The aligned score does not show a resolved advantage over random growth in any of these three tests. Its paired mean differences are $+0.000999 \pm 0.001588$ nats on the teacher task, $+0.002634 \pm 0.009912$ on the localized task, and -0.000666 ± 0.002350 on the linear-null task, where each uncertainty is a 95% Student- t interval half-width across the 32 fresh seeds. The localized task offers clear scope for improvement from additional capacity, but the aligned information score does not identify the most useful operation: its mean risk is 0.518572, compared with 0.493478 for the specified entropy proxy and 0.488745 for the fixed-width approximate-compute control. The entropy proxy has a paired advantage over random growth of 0.022459 ± 0.011681 nats on this task, whose interval excludes zero; this positive measured

Table 2: Fresh-seed operation-aligned information-score ablation. Risks integrate the teacher label probability on 8192 held-out Gaussian inputs, and are averaged across 32 seeds. The last column gives aligned-CMI minus random-growth risk with a paired 95% Student- t interval half-width. Fixed width 3 uses the leading-width compute proxy; its initialization differs from the grown model.

Task	Aligned CMI	Random	Entropy	Fixed 3	Difference $\pm$ half-width
teacher	0.40350	0.40250	0.40364	0.40093	$+0.00100 \pm 0.00159$
localized	0.51857	0.51594	0.49348	0.48874	$+0.00263 \pm 0.00991$
linear null	0.53534	0.53601	0.53724	0.53359	-0.00067 ± 0.00235

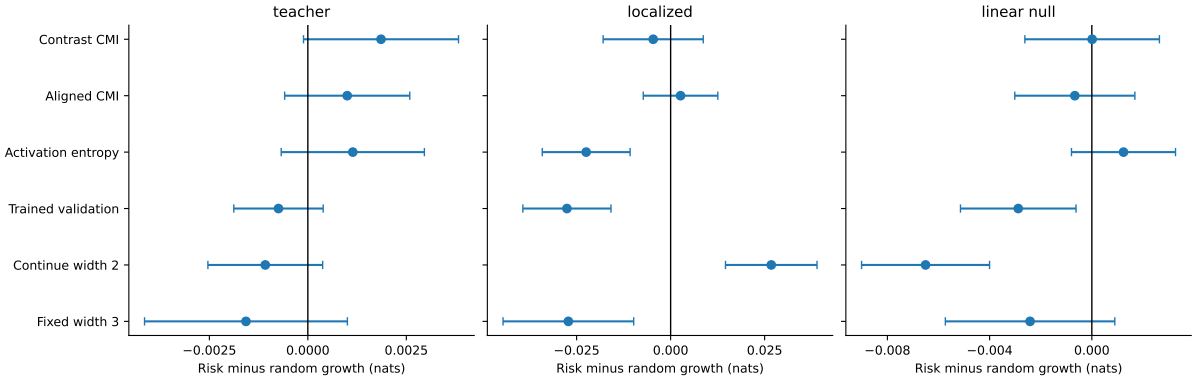


Figure 4: Fresh-seed information-score ablation: paired risk differences from random growth over 32 independent repetitions, with 95% Student- t intervals. Aligned CMI uses the exact initial symmetric split logit deformation; contrast CMI uses the original antisymmetric activation contrast. Trained validation fits all eight candidates; fixed width 3 has a distinct initialization and a leading-width compute proxy. Risks average the true conditional label loss over 8192 held-out Gaussian inputs, not an exactly enumerated population. This plot is generated solely from saved experimental outputs.

finding is retained alongside the information-score nulls. This observation does not make marginal activation entropy a valid invariant principle. It shows that, in this experiment, the proposed label-information proxy failed even to outperform a simple competing score.

On the linear-null task, continuing width two achieves mean risk 0.529499, below the aligned-growth risk 0.535339. Information-score selection in this study was forced to grow and therefore did not itself solve the decision of whether to grow. Trained validation often improves candidate selection, but it pays to fit all candidates and is not evidence for a cheap information-based controller. The original 32-seed antisymmetric-proxy run is also preserved: its apparent four-seed localized-task benefit did not become a resolved conditional-information advantage after expansion. The audit-triggered aligned ablation prevents that initially flawed proxy from carrying the conclusion.

The supported advance is consequently narrow and explicit. Input-residual geometry can identify a beneficial real neuronal operation from finite samples at a constructed stationary checkpoint, whereas unit-local scalar observables cannot identify its direction. Equal-capacity neural training verifies that this matters under a short optimization budget. Longer training, ordinary pretraining, irrelevant inputs, and no-signal tasks expose its limitations. The information-score experiments further show that defining and estimating a conditional label-information quantity does not by itself supply an effective neural structural controller. These findings establish a testable bridge and its failure boundaries, not a general autonomous architecture-learning method.

All raw runs, failed or unfavorable conditions, seed ranges, learned matrices, model parame-

ters, resource proxies, and rerunnable figure scripts are included. The main neural studies use only the existing local NumPy/SciPy environment; no GPU, cloud resource, paid API, external dataset, or system-wide installation was used. The experiments contain one structural decision per run and do not establish the behavior of repeated, lifelong growth.

8 What this advances, and what remains unsupported

Relative to Phase I, this work replaces a purely finite-representation obstruction with explicit neural parameters, finite observations, and actual learning. It does not identify neural width with the number of representation states. The observation theorem explains why a scalar information content or an activation summary can fail to determine a useful operation even when the relevant information is present in the input. The sample theorem gives a realizable context-based decision with a matching necessary sample order in that restricted problem. The singular positive split theorem shows that width alone also misses a resource: how outgoing proportions and incoming movement approach the structural change, even without negative weights or extra children.

The trainable step achieved is modest but concrete. A controller computes an empirical statistic from a unit’s input and residual, selects a real neural duplication direction, trains both resulting hidden units, and can accept or reject growth using a disjoint holdout set. The evidence establishes its short-budget benefit at a constructed stationary point and measures where that benefit disappears. It does not establish that activation entropy, conditional information, or split curvature is generally the right trigger for autonomous structure adaptation. One-event growth and rejection are implemented; a stable repeated lifetime controller is not.

Several boundaries carry the conclusions. The activation-only theorem is about a passive pre-operation observation channel; ordinary subsequent training or active input-dependent probing can break indistinguishability. The high-order proof keeps an intercept and a particular operation family fixed, and selects its direction and coefficient with population knowledge. It is not a guarantee that SGD discovers a vanishing-weight path. Its population gains can be many orders smaller than sampling error or numerical precision. The neural experiments are small synthetic tanh classifiers, not broad deep-learning benchmarks, biological neuron models, or tests of nonstationary continual learning. Fixed-width baselines often match or outperform the adaptive alternatives. Parameter-update proxies do not measure hardware FLOPs, memory, or wall-clock efficiency of a deployable controller. Holdout gating consumes another fitted branch and extra data; its normal approximation is not a theorem-level certificate.

The nearest-source review substantially limits the novelty claim: known curvature growth and known boundary escape provide the foundation. The new claims are the exact observation/task witness with quantitative evidence requirements and the constrained positive-weight joint-limit separation, together with openly adverse implementation evidence. Their equivalence to a broader existing formulation remains a question for external review. The project supplies complete proofs, independent checks and raw results, but does not claim external peer review or publication readiness.

9 Reproducibility and research record

All Phase II materials are under `phase2/`. The top-level Phase I manuscript and outputs remain unchanged. `README.md` gives exact commands; `code/reproduce.py` regenerates the performed computations, figures and paper. Raw outputs distinguish exploratory pilots, the expanded probe, the new operation-aligned evaluation, the curvature bridge, population calculations and sample-only tests. Saved seeds and deterministic generators reconstruct all data. The neural bridge saves event directions, matrices, acceptance decisions, post-operation risks and

parameters. It does not pretend to log every optimizer iterate.

The runtime is local macOS arm64, Python 3.12.14, NumPy 2.3.5, SciPy 1.18.1, mpmath 1.3.0 and Matplotlib 3.11.2, with eight reported logical CPUs. NumPy CPU training avoids a missing PyTorch dependency. No paid API, remote GPU, publishing service, private data upload, or system-wide installation was used. LaTeX/Poppler compile and render the companion PDF. A concise status file and contribution/evidence ledger record completed work, corrections, objections and remaining gaps.

Independent numerical implementations cross-check the proof calculations and learning routines. The high-order direction was revised after counterexample search, and the information-probe mismatch was corrected by an additional operation-aligned evaluation with new seeds. These historical checks do not constitute external peer review.

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